Secure Document Access in Power BI – Proposed Solution

🧠 1. PROBLEM OVERVIEW

The objective is to enable users to securely access and view documents, such as PDFs, directly within a Power BI dashboard.

The key challenge is to ensure that:

- Only authorized users can access specific documents
- Sensitive data is protected from unauthorized access
- The solution remains scalable and easy to manage

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2. PROPOSED SOLUTION

The proposed solution is to store documents in a secure external storage system, such as **Azure Blob Storage or SharePoint**.

Instead of storing documents directly in Power BI, the dashboard will display **secure links** to these documents.

Access to these links will be controlled through:

- User authentication
- Role-based access control
- Dynamic filtering within Power BI

This ensures that users only see and access documents they are permitted to view.

For example, an enterprise user logs into the dashboard and can access only their company-specific reports stored in SharePoint. A free user, on the other hand, will

only see limited documents based on their access level. This ensures that each user views only relevant and authorized content.

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3. ARCHITECTURE DESIGN

The system works as follows:

1. The user logs into Power BI using secure authentication (Azure Active Directory)
2. Power BI identifies the user and their role
3. Row-Level Security (RLS) is applied to filter accessible data
4. The dashboard displays only the documents allowed for that user
5. When a user clicks a document link, it opens securely from external storage

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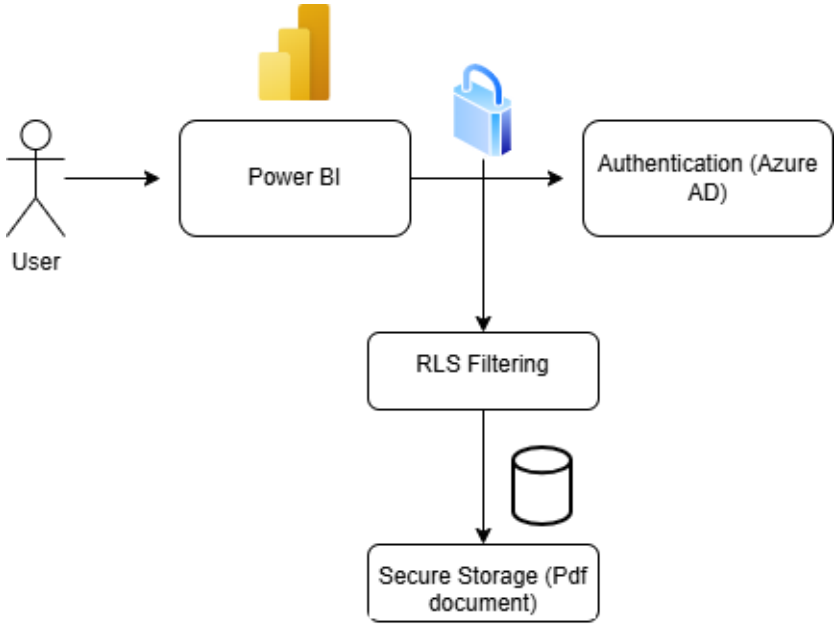
🔁 Simple Flow:



🔐 4. SECURITY MECHANISMS

The solution ensures security through multiple layers:

🔑 Authentication

- Users log in through Azure Active Directory
- Ensures identity verification

🔒 Authorization

- Row-Level Security (RLS) ensures users only see allowed documents

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📁 Secure Storage

- Documents are stored in secure cloud systems (Azure Blob / SharePoint)
- Access permissions are strictly controlled

⏳ Optional: Expiring Links

- Temporary or token-based URLs can be used
- Prevents unauthorized sharing of links

⚠️ 5. CHALLENGES

Some challenges in this approach include:

- Managing access control for a large number of users
- Ensuring links are not shared outside authorized users
- Handling latency when loading documents from external storage
- Maintaining synchronization between Power BI and document storage

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⚖️ 6. TRADE-OFFS

Aspect	Advantage	Limitation
Security	Strong access control	More complex setup
Scalability	Easy to scale with cloud storage	Dependency on external systems
User Experience	Seamless integration	Slight delay in document loading

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🧠 7. CONCLUSION

This solution provides a secure and scalable way to enable document access within Power BI.

By combining external storage, authentication, and row-level security, it ensures that:

- Users only access authorized documents
- Sensitive information remains protected
- The system remains flexible and scalable